

Final Evolution Lab — Manual AWS Provisioning Guide

Prerequisites

- AWS Account with access to G5 instances
- EC2 Key Pair created in your target region
- Basic familiarity with AWS Console

Step 1: Launch EC2 Instance

1. Go to **EC2 → Launch Instance**
2. Configure:
 - **Name:** FinalEvolutionLab-Build
 - **AMI:** Ubuntu Server 22.04 LTS (64-bit x86)
 - **Instance type:** g5.2xlarge (1 NVIDIA A10G GPU, 8 vCPU, 32GB RAM)
 - **Key pair:** Select your key pair
 - **Network:**
 - Create new VPC or use default
 - Enable “Auto-assign public IP”
 - **Storage:** 500 GB gp3 (IOPS: 6000, Throughput: 400 MB/s)
 - **Security Group:** Create new with these rules:

Type	Port Range	Source	Description
SSH	22	Your IP	SSH access
HTTP	80	0.0.0.0/0	Website
HTTPS	443	0.0.0.0/0	SSL
Custom	3000	0.0.0.0/0	Web Frontend
Custom	8888-8889	0.0.0.0/0	Pixel Streaming
Custom	3478	0.0.0.0/0	TURN TCP
UDP	3478	0.0.0.0/0	TURN UDP
UDP	49152-49200	0.0.0.0/0	Media Relay

1. **Launch Instance**

Step 2: Allocate Elastic IP

1. Go to **EC2 → Elastic IPs → Allocate**
2. Associate with your instance
3. Note the IP: _____

Step 3: Create IAM Role (Optional)

1. Go to **IAM → Roles → Create Role**
2. Trusted entity: EC2
3. Attach policies:
 - AmazonS3FullAccess
 - CloudWatchAgentServerPolicy
4. Name: FEL-InstanceRole
5. Attach to instance: EC2 → Instance → Actions → Security → Modify IAM role

Step 4: Connect & Setup

```
# SSH into instance
ssh -i your-key.pem ubuntu@YOUR_ELASTIC_IP

# Transfer project (from your local machine)
scp -r -i your-key.pem /path/to/rork-final-evolution-lab ubuntu@YOUR_ELASTIC_IP:/home/ubuntu/

# Or clone from GitHub
export GITHUB_TOKEN="ghp_your_token"
git clone https://[${GITHUB_TOKEN}]@github.com/finalevolutiongroup/final-evolution-lab.git rork-final-evolution-lab

# Run bootstrap
cd /home/ubuntu/rork-final-evolution-lab
chmod +x deploy/aws/bootstrap.sh
./deploy/aws/bootstrap.sh

# After bootstrap completes, run the pipeline
./deploy/aws/deploy_final_evolution_lab.sh --build --deploy
```

Step 5: Transfer Project from Current VM

If you have the project on another machine:

```
# Option A: rsync (fastest, supports resume)
rsync -avz --progress -e 'ssh -i key.pem' \
/home/ubuntu/rork-final-evolution-lab \
ubuntu@YOUR_IP:/home/ubuntu/

# Option B: tar + scp
cd /home/ubuntu
tar czf fel-project.tar.gz rork-final-evolution-lab \
--exclude='node_modules' --exclude='.git' --exclude='Builds'
scp -i key.pem fel-project.tar.gz ubuntu@YOUR_IP:/home/ubuntu/
# Then on remote: tar xzf fel-project.tar.gz

# Option C: S3 intermediate
aws s3 cp --recursive rork-final-evolution-lab s3://your-bucket/fel/
# On remote:
aws s3 cp --recursive s3://your-bucket/fel/ rork-final-evolution-lab
```

Step 6: Set Credentials

```
# Edit .env file
nano /home/ubuntu/rork-final-evolution-lab/.env

# Required:
GITHUB_TOKEN=ghp_...           # Epic Games GitHub PAT
MESHY_API_KEY=msy_...           # Meshy AI key
LUMA_API_KEY=...                # Luma AI key

# Optional:
GAIA_API_KEY=...                # Gaia API
RUNWAY_API_KEY=...              # Runway API
STABILITY_API_KEY=...           # Stability AI
NOTIFICATION_EMAIL=you@...      # Email alerts
SLACK_WEBHOOK=https://hooks...  # Slack alerts
```

Step 7: Run Pipeline

```
# Interactive mode (recommended for first run)
./deploy/aws/deploy_final_evolution_lab.sh --interactive

# Or direct
./deploy/aws/deploy_final_evolution_lab.sh --build --deploy

# Monitor progress
./deploy/aws/monitor_dashboard.sh --watch
```

Cost Estimate

Resource	Cost/Hour	Monthly (24/7)
g5.2xlarge	~\$1.21	~\$871
500GB gp3 EBS	~\$0.055/h	~\$40
Elastic IP	Free	Free (in use)
Data Transfer	~\$0.09/GB	Varies
Total Build	~\$1.27/h	-

Tip: Build takes ~4-8 hours. Stop instance when not in use!

```
# Stop to save money (preserves EBS)
aws ec2 stop-instances --instance-ids i-xxx
# Start when needed
aws ec2 start-instances --instance-ids i-xxx
```